Introduction

Oncosense is a wearable armband device designed for non-invasive optical analysis of the skin. The prototype integrates an 850 nm near-infrared (NIR) LED as a light source, an Adafruit spectrometer module as the detector, and an Adafruit Qt Py microcontroller for control and data acquisition. The LED and spectrometer are mounted facing the skin, while the microcontroller sits on the outer side of the band. The original motivation for Oncosense was to leverage advanced optical phenomena – including linear techniques like diffuse reflectance and fluorescence, as well as non-linear techniques like second harmonic generation (SHG) – to distinguish carcinogenic skin cells from normal tissue. In theory, the LED would excite endogenous molecules in skin lesions (or exogenous contrast agents), and the spectrometer would detect unique spectral signatures to identify cancerous changes. This approach is grounded in a rich literature of optical diagnostics: for several decades researchers have explored optical methods for early skin cancer detection, given skin's easy accessibility and the high incidence of skin tumors. However, implementing complex nonlinear optics in a compact, wearable form is challenging. Thus, the current iteration of Oncosense has pivoted to a simpler but still valuable application: using photoluminescence and diffuse reflectance from the skin to assess skin hydration levels. By measuring how the skin's reflectance at 850 nm changes (for example, before and after applying moisturizers or sunscreen), the device can infer the hydration or moisture content of the stratum corneum. This literature review will survey the background and state-of-the-art relevant to Oncosense – from historical developments in optical skin diagnostics (both cancer detection and hydration sensing) to specifics of each hardware component – and highlight recent breakthroughs that enable a wearable spectroscopic sensor for skin health monitoring.

Optical Techniques for Skin Cancer Detection

Optical sensing and imaging have long been investigated as non-invasive alternatives or complements to biopsy for diagnosing skin cancer. Two broad categories of optical techniques have shown promise: **diffuse reflectance spectroscopy (DRS)** and **fluorescence spectroscopy**, including autofluorescence. DRS involves shining light (often visible or NIR wavelengths) on the skin and analyzing the reflected light spectrum. This yields information about tissue morphology and composition – for example, hemoglobin absorption features and melanin content affect the reflectance spectrum. **Autofluorescence (also called light-induced autofluorescence spectroscopy, LIAFS)** entails exciting the skin with UV or blue light and detecting the faint fluorescence emitted by endogenous molecules. This can reveal biochemical changes, since many skin components (collagen, elastin, NADH, flavins, etc.) naturally fluoresce and their concentrations or states change in dysplasia or malignancy. Each technique alone provides useful but partial information; reflectance mainly highlights structural and pigment differences, whereas fluorescence highlights metabolic or biochemical differences. Notably, combining the two can significantly improve diagnostic accuracy. In a clinical study of a hybrid spectral system, Borisova et al. showed that using both autofluorescence and reflectance data yielded

higher sensitivity and specificity for skin cancer than either method alone. For example, basal cell carcinoma (BCC) lesions consistently showed much lower autofluorescence intensity than surrounding normal skin, while squamous cell carcinomas (SCC) showed higher fluorescence than normal skin. Malignant melanomas tended to exhibit extremely weak fluorescence (due to strong absorption by melanin and loss of certain fluorophores), making them distinguishable from benign moles by their suppressed LIAFS signal. By incorporating reflectance features (such as hemoglobin absorption dips and overall shape of the continuum) alongside fluorescence intensity, the researchers could improve melanoma diagnostic accuracy to over 90% – in one report, a combined autofluorescence+DRS approach identified malignant melanoma with about 93% accuracy. These results underscore that multi-spectral optical sensing can detect the subtle differences between cancerous and healthy skin tissues.

Another avenue of optical diagnostics involves advanced microscopy techniques, particularly **non-linear optical imaging**. Second harmonic generation (SHG) microscopy and two-photon excited fluorescence are examples that have been applied in laboratory research on skin cancer. SHG is a nonlinear phenomenon where certain structured molecules (those lacking inversion symmetry) generate light at exactly half the excitation wavelength when excited by a high-intensity pulsed laser. In biological tissues, collagen is a primary source of SHG signal (collagen fibers are non-centrosymmetric and produce strong SHG). SHG imaging has provided valuable insights into the tumor microenvironment, since tumors often disrupt the collagen architecture in the dermis. For instance, SHG microscopy can visualize tumor-associated collagen changes – fibers becoming disordered or entangled – without any staining or labels. This label-free specificity to collagen is a major advantage of SHG, and indeed researchers have found that SHG can differentiate cancerous tissue from normal by the altered collagen patterns in and around tumors. For example, polarisation-sensitive SHG has been used to distinguish diseased/cancerous skin from healthy skin by quantifying changes in collagen fiber orientation and density. However, these nonlinear techniques typically require femtosecond lasers and high numerical aperture optics, making them expensive and not yet portable. They are more suited to microscopic imaging or ex vivo analysis (or high-end in vivo microscopes) than to a wearable device. Still, the knowledge gained from SHG studies – that collagen structure is a key differentiator in cancers – complements the macro-scale information from DRS and fluorescence.

In summary, over years of development, optical methods ranging from straightforward reflectance spectroscopy to sophisticated multiphoton microscopy have demonstrated the ability to detect skin cancer signatures. *Linear* optical techniques (DRS and autofluorescence) have even been tested in vivo in clinical settings, showing high accuracy for certain lesion types when multiple spectral features are combined. These could potentially be miniaturized into screening tools. *Non-linear* techniques like SHG offer deeper insights at the microscopic level (e.g. collagen alterations), but are currently confined to research labs. The Oncosense project initially aspired to incorporate both linear and nonlinear optics – for example, using the 850 nm LED to excite fluorescence or even attempt SHG (though true SHG would require a different light source) – in order to discriminate cancerous cells. While that full vision is ambitious for an armband, it is grounded in the demonstrated optical contrasts between healthy and malignant skin.

Skin Hydration Measurement by Optical Methods

As Oncosense's focus shifted to assessing skin hydration via optical reflectance, it taps into another rich area of skin research. Skin hydration (primarily the water content in the stratum corneum and epidermis) is crucial for skin health and appearance. Traditionally, hydration is measured by electrical methods (like capacitance or conductance-based probes), but these can be influenced by environmental factors and only probe very superficial layers. Optical techniques offer a non-contact, rapid means to gauge hydration changes throughout the skin depth. Water has characteristic absorption bands in the near-infrared: notably around 970 nm, 1450 nm, and 1950 nm (due to O–H vibrational overtones). Diffuse reflectance spectroscopy in the 400–1000 nm range can capture changes in both absorption and scattering that occur as the skin's hydration state changes. For instance, a recent study by Chen *et al.* (2023) demonstrated that using DRS from 400–1000 nm to monitor hydration is effective and correlates strongly with standard capacitance measurements (correlation coefficient ~0.93 between optically derived absorption coefficient and skin capacitance). The optical method could clearly detect when skin hydration was enhanced by external means – for example, after applying soaked hydrating pads or cosmetic facial masks, the derived optical parameters changed significantly, indicating increased water content. The advantages of such an optical approach include its non-invasive nature, speed, and suitability for continuous monitoring.

Interestingly, the interaction of light with skin is markedly affected by hydration because water and hydration-induced structural changes alter both **absorption** and **scattering**. A well-moisturized stratum corneum has a lower refractive index mismatch with the underlying tissue (since water fills air gaps), thus it scatters light less and allows deeper penetration. Empirical studies confirm this: Jiang *et al.* (2011) found that when skin was dehydrated (e.g. by applying acetone/alcohol), its diffuse reflectance spectrum became brighter (higher intensity) and the derived scattering coefficient increased, whereas hydrated skin (after applying moisturizers) showed reduced reflectance and scattering. In essence, *dry skin reflects more light diffusely and appears whiter or duller, while hydrated skin absorbs and transmits more, appearing darker but more translucent*. Hydrated skin also exhibits increased translucency (light can penetrate to deeper layers) because surface scattering is suppressed. This optical effect is one reason why applying a moisturizer can visibly change skin appearance (more glow or pinkish tone as underlying blood shows through). For Oncosense, which uses an 850 nm LED, we note that 850 nm lies just beyond the visible range in a region where water absorption is not at its peak but melanin and hemoglobin absorption are tapering off. Changes in skin hydration at 850 nm will primarily manifest as changes in scattering (since 850 nm light can penetrate into the dermis and its return signal will be modulated by how much scattering occurs in the stratum corneum and epidermis). A freshly moisturized skin patch is expected to reflect *less* diffuse light at 850 nm compared to the same patch when very dry, due to the reduced scattering at the surface – unless there is a strong specular reflection from a wet surface layer, which can be mitigated by measurement geometry.

Researchers have also developed dedicated **multi-wavelength optical sensors for skin hydration**. One example is the sensor by Mamouei *et al.* (2021), which used three NIR LED wavelengths (around 970, 1200,

and 1450 nm) and photodetectors to continuously monitor dermal water content. This device separated the source and detector by a small distance to sense reflectance from a fixed depth, and it incorporated onboard amplification and analog-to-digital conversion for portability. The design comprised a probe module (with the LEDs at those wavelengths and a photodiode detector, plus amplifiers) and a main electronics module (with microcontroller, ADC, and power supply). By comparing the sensor's readings to a laboratory spectrophotometer and gravimetric water measurements in skin samples, the authors showed the wearable sensor could accurately track changes in skin water content over time. This validates the principle that even a few discrete NIR wavelengths are enough to monitor hydration because water's absorption features and the overall optical properties shift noticeably as hydration changes.

Figure: An example of a multi-wavelength skin hydration sensor. *(a)* Block diagram of the system with a probe containing LEDs and photodetectors and a main electronics module; *(b)* the physical probe next to a coin for scale. Such devices use optical reflectance at NIR wavelengths to quantify skin water content.

The Oncosense device, although currently using only a single NIR LED (850 nm) and a spectrometer covering roughly 600–900 nm, operates on the same basic concept. By calibrating the intensity of reflected light (or spectral shape) against known hydrated and dehydrated conditions, it can serve as a crude *optical corneometer*. For instance, if Oncosense records a higher reflectance at 850 nm from a skin site after it has been left to dry out (or cleansed with alcohol) and a lower reflectance after a moisturizer is applied, this difference can be quantitatively related to the change in skin hydration. The literature assures that these differences are real and measurable: diffuse reflectance increased by dehydration and decreased by moisturization has been documented both in vivo and in controlled optical experiments. Thus, even though the initial lofty goal of detecting cancer via fluorescence/SHG is on hold, the current hydration-sensing application of Oncosense is backed by solid scientific precedent and addresses an important aspect of skincare.

Key Components and Historical Development

To understand how a device like Oncosense is possible, it is useful to review the components involved and how each has evolved in the context of biomedical sensing:

- **Near-Infrared LED (850 nm):** Light-emitting diodes at specific wavelengths have become ubiquitous, but it's noteworthy that 850 nm LEDs are widely used in electronics (e.g. remote controls, IR illumination for cameras) and biomedical optics. The choice of 850 nm is partly practical – silicon photodiodes and many spectrometers have good sensitivity up to ~850–900 nm, and there exists a "therapeutic/window" region in red/NIR light where penetration into tissue is maximized due to lower absorption by blood. In skin, 850 nm light penetrates a few millimeters, reaching the dermis, but melanin still absorbs in this range (especially for darker skin, melanin absorption causes a monotonic decrease in transmittance across 650–850 nm). Historically, dual-wavelength optical probes for blood oximetry used a similar IR wavelength (~880–940 nm) along with a red light to measure oxygen

saturation; this established that IR light can probe perfused tissue noninvasively. Modern 850 nm LEDs are high-powered and stable, suitable for sensing applications where consistent illumination is needed. Interestingly, there is even research on using 830–850 nm LEDs therapeutically (photobiomodulation) to influence skin pigmentation – some studies indicate these wavelengths can modulate melanin production. For Oncosense, the LED's job is simply to provide a controlled light input. Because it is in the NIR, it will also cause any **photoluminescence** in that range to be emitted from the skin (though human skin's significant autofluorescence is usually in visible range under UV excitation, not under 850 nm). If any fluorescent moisturizers or skincare ingredients are present that emit in NIR, the spectrometer could in principle detect that as well, though this is a secondary consideration.

- **Miniature Spectrometer (Adafruit module):** The spectrometer component in Oncosense is likely based on a chip like the AMS **AS7263** 6-channel spectral sensor or a similar multispectral sensor. Adafruit's "spectrometer" breakout boards typically use such integrated sensors rather than a prism/grating spectrometer. The AS7263, for example, provides readings at six discrete NIR bands centered at approximately 610, 680, 730, 760, 810, and 860 nm, each with about 20–40 nm bandwidth. Its sister chip AS7262 covers six visible bands from violet to red. These sensors use on-chip interference filters to achieve spectroscopic discrimination in a very tiny package (a few mm in size). Historically, achieving spectroscopy required large diffraction gratings or scanning monochromators. But the recent development of solid-state spectral sensors has revolutionized portable spectroscopy. As Workman (2024) notes, **low-cost (< $100) multispectral sensors** in the Vis/NIR are now available and can be embedded in consumer electronics and wearables. This is exactly the category into which the Adafruit spectrometer falls – it's affordable and small enough to integrate into an armband. While such sensors don't produce a full high-resolution spectrum, they are sufficient for many sensing tasks (like color measurement, or detecting broad absorption features). In Oncosense's case, the spectrometer can measure how the intensity of reflected light is distributed across those six channels. If the device is used for hydration, one might primarily look at the overall intensity at 810 or 860 nm channels. If extended to skin lesion detection, the relative differences between channels (say 680 nm vs 810 nm reflectance, which could relate to hemoglobin vs water content) could be features of interest. The **historical development** of mini-spectrometers has been rapid: from the first micro-patterned filters on chips in the 2010s to today's **MEMS based spectrometers** and even computational spectrometers the size of a grain of rice. This trend is what makes Oncosense feasible – a decade or two ago, one could not imagine a full spectroscopic setup on a 3-cm wristband.

- **Microcontroller (Adafruit Qt Py):** The Qt Py is a tiny microcontroller board (about the size of a postage stamp) that can be based on various MCU chips (often ATSAMD21, RP2040, etc.) and is used to interface sensors and communicate data, typically via USB or wireless if paired with a radio. The role of the microcontroller is to drive the LED (possibly modulating it or turning on/off), to read data from the spectrometer (usually via I^2C or SPI communication), and to process or transmit that data to a higher-level system (like a PC or smartphone). In the broader context, microcontrollers have enabled the

explosion of **wearable health tech** by providing computing power in a very small power-efficient package. Whether it's a Fitbit or a medical patch, an MCU collects sensor data and runs algorithms. The Qt Py's placement "away from the skin" in the armband is simply to avoid the bulk and any heat near the sensing area. Historically, the miniaturization of microcontrollers (and the increase in their performance) means that even complex tasks like running machine learning models on sensor data can be done on-device. This could be relevant if Oncosense evolves to use machine learning for pattern recognition (for instance, distinguishing a malignant lesion's spectral signature from a benign one). It's worth noting how companies like Google and Apple are leveraging optical sensors in wearables: for example, the latest smartwatches use optical heart rate sensors (green and IR LEDs with photodiodes) and even SpO_2 sensors (red/IR) to monitor vital signs. Those are essentially simplified spectroscopic instruments (two wavelengths). As we move forward, it's expected that more advanced spectroscopy (with multiple wavelengths, perhaps even including UV or longer IR) will be incorporated into consumer wearables for things like hydration, skin condition, or analyte sensing.

- **Nonlinear Optics Components (future consideration):** While not in the current prototype, if one were to consider SHG or fluorescence seriously, additional components would be needed: e.g. a UV LED (around 365–405 nm) to excite autofluorescence as was done by some researchers, or a pulsed laser for SHG. UV LEDs are now commonly available and could be integrated to induce fluorescence of skin (some devices like the Dermalite Handyscope use UV to visualize skin fluorophores). SHG, however, would likely remain beyond reach for a battery-powered wearable, because it requires both a high peak power laser and detection of a specific second-harmonic wavelength (for collagen, typically using a 800 nm laser to get 400 nm SHG signal, which would be absorbed in skin quickly and not make it out easily). It's more plausible that Oncosense or its successors could implement **two-photon fluorescence** using NIR pulses to excite UV-range fluorescence in tissue – but again that demands a mode-locked laser. For now, the literature knowledge of nonlinear optics in skin is mainly inspirational for the project rather than directly implementable hardware.

Recent Breakthroughs and Future Directions

The convergence of the above technologies has led to a number of recent breakthroughs that are relevant to Oncosense's mission. As mentioned, miniaturized spectrometers are now a commercial reality, and research summaries predict they will soon be ubiquitous in consumer devices. In the realm of skin health diagnostics, we are seeing:

- **Portable skin imaging/spectroscopy devices:** e.g. devices combining smartphone cameras with multispectral lights to do rudimentary skin lesion analysis, or handheld Raman spectroscopy units for skin cancer detection. While Raman is another spectroscopic technique (based on inelastic scattering), it has been used experimentally for melanoma diagnosis with good accuracy, and miniaturized Raman

spectrometers are emerging. A combination of Raman and autofluorescence in a handheld probe was even shown to identify basal cell carcinoma during surgery. These show that optical spectroscopic diagnosis is moving from bench to bedside.

- **Wearable health monitors using NIR:** Beyond heart rate, there are wearables for things like muscle oxygenation (using NIR spectroscopy of hemoglobin) in athletes. Also, a wearable fNIRS (functional NIR spectroscopy) headband was demonstrated for brain monitoring, using multiple IR light sources and detectors in a head-mounted unit. This illustrates that even multi-channel NIR measurement of tissue can be made wireless and wearable, a concept very aligned with Oncosense (simply applied to skin instead of brain).

- **Skin analytics in cosmetics:** Cosmetics companies and dermatology researchers are very interested in quantifying skin properties (hydration, pigmentation, elasticity) objectively. Studies like Qassem & Kyriacou (2014–2015) used near-infrared reflectance to monitor how daily moisturizer use changes skin optical properties over weeks. There are even smartphone-linked gadgets that claim to measure skin moisture optically. The trend suggests that by presenting the Oncosense device to a place like MIT, one could spark interest in further developing it into a practical tool for skincare or cancer pre-screening.

In summary, the literature and current state-of-art strongly support the components and approaches in the Oncosense project. Optical techniques offer a spectrum (pun intended) of possibilities for non-invasive sensing: from detecting life-threatening conditions like melanoma by its optical signature, to something as routine as checking if your skin is dry and needs moisturizer. Oncosense's initial grand vision to differentiate cancerous cells via SHG and fluorescence was certainly ambitious – leveraging the known differences in fluorophore content and collagen structure in tumors – but it faced practical limitations. By refocusing on photoluminescence and reflectance for hydration, the project is targeting a more immediate and demonstrable goal, which also serves as a stepping stone. A device that reliably measures skin hydration optically could itself be quite valuable (for skincare research or consumer use), and the experience gained can inform future iterations that might reincorporate additional wavelengths or modalities for broader skin health diagnostics. The key enablers are all trending in the right direction: LEDs are available in many wavelengths (including deep UV and IR), tiny spectrometers and photodiode arrays can cover from UV to mid-IR, and powerful microcontrollers can run sophisticated analyses on the fly. As the literature shows, combining multiple optical signals yields the best results, so a logical future step for Oncosense would be to add more LEDs (e.g. a UV LED for autofluorescence excitation, a second IR LED around 940 nm or 1450 nm for water absorption) and perhaps use machine learning on the spectrometer data to detect patterns. With interdisciplinary efforts – drawing on photonics, biomedical engineering, and machine learning – Oncosense and devices like it could eventually provide wearable, real-time sensing for early detection of skin cancers or continuous monitoring of skin condition, fulfilling a vision that pioneers in optical diagnostics have been approaching for decades.

Conclusion

This literature review highlighted how the Oncosense project builds upon a foundation of optical biomedical research. From the early experiments of differentiating tumors by their fluorescence and reflectance, to the latest miniaturized sensors for skin hydration and vital signs, the trajectory is clear: optical sensing is moving towards portability and wearability. Each component of Oncosense – NIR LED, spectral sensor, microcontroller – has benefited from technological advances that made it smaller, cheaper, and more capable, enabling an undergraduate project to attempt tasks that once required a full lab. The review also underscores an important principle: **multimodal optical analysis** (using multiple wavelengths or multiple techniques) is often necessary to get reliable diagnostic information. Even though Oncosense's current hydration demo uses a single wavelength reflectance, the ultimate power of such a device will likely come from integrating more optical channels. As presented to peers at MIT, Oncosense can be seen as a case study at the intersection of photonics and health: it leverages known optical biomarkers of skin state (hydration or even malignancy) and employs cutting-edge but accessible hardware to detect them. Continuing this project will mean not only refining the hardware (perhaps adding a broader spectral range or stronger light source) but also validating it against gold-standard measurements (e.g. comparing its readings to clinical instruments or biopsy results, in the case of cancer detection attempts). The literature provides both inspiration and caution – optical methods are sensitive and can be highly accurate, but biological variability (skin type, ambient light, etc.) must be accounted for. By referencing peer-reviewed studies and breakthroughs throughout this review, we have ensured that the Oncosense project is guided by scientific evidence. In doing so, we pave the way for it to mature from a prototype gadget into a truly impactful tool, whether in dermatology clinics or in everyday skincare routines.

Sources: The information above was synthesized from numerous peer-reviewed articles and reviews. For instance, Borisova *et al.* (2014) demonstrated the combined use of autofluorescence and diffuse reflectance for skin cancer diagnostics. Jiang *et al.* (2011) examined optical changes due to skin moisturization, confirming decreased scattering and reflectance with increased hydration. Chen *et al.* (2023) showed the efficacy of 400–1000 nm DRS for quantifying skin hydration, with strong correlation to conventional measures. The design of wearable optical sensors for dermal water content was detailed by Mamouei *et al.* (2021), and the capabilities of the AS726x spectral sensors are documented in hobbyist literature (Dr. H's Instructables, 2018). Broader context on portable spectroscopy and its future was drawn from Workman (2024). Second harmonic generation's role in collagen imaging was discussed by Park *et al.* and others. These and other references (see inline citations) collectively underpin the concepts and facts presented in this review, illustrating the continuity from historical developments to the Oncosense project's implementation.